

Channel State Information in Near-Field Cell-Free XL-MIMO

Perfect CSI vs Estimated CSI: how each is computed, why, and with which references

Purpose. This note explains, in plain language and with the governing equations, the two channel-state-information (CSI) regimes used in the cell-free hybrid near-field / far-field (NF/FF) XL-MIMO uplink: (i) **perfect CSI** (a genie upper bound) and (ii) **estimated CSI** (pilot-based LMMSE). For each, it states how the channel is obtained, why that choice is made, who established the method, and which reference supports it — separately for the far-field and the near-field case. It also assesses what the current code actually does and flags one inconsistency to correct in the new paper.

1. The channel that CSI refers to

Each AP l has a uniform linear array of N antennas at half-wavelength spacing ($\Delta = \lambda/2$). Element n of AP l sits at position $\mathbf{r}_{\{n,l\}}$. A single-antenna user k at position $\mathbf{u}_k$ has, to AP l , the channel vector $\mathbf{h}_{\{l,k\}}$ in $\mathbb{C}^N$. Two regimes are distinguished by the AP–user distance $d_{\{l,k\}}$ relative to the Rayleigh (Fraunhofer) distance $d_{\text{Ray}} = 2 D^2 / \lambda$, with aperture $D = (N-1) \lambda/2$:

- Far field ($d_{\{l,k\}} > d_{\text{Ray}}$): planar wavefront. The channel is modelled as correlated Rayleigh fading,

$$\mathbf{h}_{\{l,k\}} \sim \text{CN}(\mathbf{0}, \mathbf{R}_{\{l,k\}}),$$

where $\mathbf{R}_{\{l,k\}}$ in $\mathbb{C}^{N \times N}$ is the spatial correlation matrix set by the large-scale fading $\beta_{\{l,k\}}$ and the angular spread (the model of Demir–Bjornson–Sanguinetti [2]).

- Near field ($d_{\{l,k\}} \leq d_{\text{Ray}}$): spherical wavefront. Using the non-uniform spherical-wave (NUSW) model, the line-of-sight channel is deterministic given the geometry:

$$[\mathbf{h}_{\{l,k\}}]_n = \sqrt{\beta_{\{l,k\}}} \cdot \left(d_{\{l,k\}} / d_{\{n,l,k\}} \right) \cdot \exp(-j 2 \pi d_{\{n,l,k\}} / \lambda),$$

with $d_{\{n,l,k\}} = \|\mathbf{r}_{\{n,l\}} - \mathbf{u}_k\|$ the exact element-to-user distance. The amplitude term d/d_n and the non-linear phase are what distinguish the near field from the plane-wave model [4,5].

Why this matters for CSI. Perfect CSI means the receiver knows $\mathbf{h}_{\{l,k\}}$ exactly; estimated CSI means it forms an approximation $\hat{\mathbf{h}}_{\{l,k\}}$ from pilots. The two cases are computed very differently in the near field than in the far field, which is the whole subject of this note.

2. Perfect CSI (the genie upper bound)

What it is. Perfect CSI assumes the receiver knows the true channel realization $\mathbf{h}_{\{l,k\}}$ for every AP–user pair, with no estimation error. It is not ‘computed’ by an algorithm: the exact channel vector generated by the model is fed directly into the combiner. In the code this means the local MMSE filter

$$\mathbf{v}_{\{l,k\}} = \sqrt{P} \left(\mathbf{P} \sum_j \mathbf{h}_{\{l,j\}} \mathbf{h}_{\{l,j\}}^H + \sigma^2 \mathbf{I} \right)^{-1} \mathbf{h}_{\{l,k\}}$$

is built from the TRUE $\mathbf{h}_{\{l,k\}}$ (not an estimate), and the estimation-error covariance is set to zero, $\mathbf{C}_{\{l,k\}} = \mathbf{0}$.

How it is obtained in each regime.

- **Far field:** the true realization $\mathbf{h}_{\{l,k\}} \sim \text{CN}(\mathbf{0}, \mathbf{R}_{\{l,k\}})$ drawn by the simulator is used as-is.
- **Near field (cell-free):** the true NUSW vector above — computed from the exact AP–element and user coordinates — is used as-is. Because the near-field LoS channel is deterministic given geometry, ‘perfect near-field CSI’ is simply this exact spherical-wave vector.

Why it is used. Perfect CSI is the standard genie benchmark in detection papers: it isolates the detector's performance from channel-acquisition error and gives an achievable upper bound. Any estimated-CSI curve must lie below it. This is exactly the assumption your submitted paper states ('Perfect CSI is assumed at the receiver').

Who does it this way / references. Perfect-CSI benchmarking is universal; for cell-free specifically it is the upper-bound convention in Demir–Bjornson–Sanguinetti [2] and Bjornson–Sanguinetti [3]. It is a modelling assumption, not a contribution, so it needs only a standard citation.

Honest point. Perfect CSI is legitimately an UPPER BOUND: it is guaranteed to be at least as good as any estimated-CSI result, because the estimated case runs the identical detector on a noisier channel with a positive error covariance. So 'perfect $\geq$ estimated' is physics, not a tuned outcome.

3. Estimated CSI: the far-field (LMMSE) case

Idea in one line. Users send known pilot sequences; each AP correlates the received pilot with the known sequence and applies a linear minimum-mean-square-error (LMMSE) filter to turn the noisy observation into the best linear estimate of the channel.

Step 1 — pilot transmission. With τ_p orthogonal pilots and a pilot-sharing set P_t (users assigned pilot t), the de-spread pilot observation at AP l is

$$y^p_{l,t} = \sqrt{P} \tau_p \sum_{i \in P_t} h_{l,i} + n_{l,t}, \quad n_{l,t} \sim \text{CN}(0, \sigma^2 \tau_p^{-1}).$$

Step 2 — LMMSE estimate. The MMSE estimate of $h_{l,k}$ and its error covariance are

$$\begin{aligned} \hat{h}_{l,k} &= \sqrt{P} R_{l,k} \Psi_{l,t}^{-1} y^p_{l,t}, \\ \Psi_{l,t} &= P \tau_p \sum_{i \in P_t} R_{l,i} + \sigma^2 I, \\ C_{l,k} &= R_{l,k} - P \tau_p R_{l,k} \Psi_{l,t}^{-1} R_{l,k}. \end{aligned}$$

By the orthogonality principle $h_{l,k} = \hat{h}_{l,k} + \tilde{h}_{l,k}$ with the estimate $\hat{h}_{l,k}$ and the error $\tilde{h}_{l,k}$ uncorrelated; $C_{l,k}$ is the covariance of $\tilde{h}_{l,k}$. The detector then uses $\hat{h}_{l,k}$ in place of h in the combiner and carries C as the residual-error term.

Pilot contamination. When several users share pilot t ($\tau_p < K$), $\Psi_{l,t}$ contains their correlation matrices, so the estimate of user k is corrupted by its co-pilot users. This is the fundamental limit of pilot-based estimation and is what your code's contamination mode ($\tau_p = K/2$) reproduces.

Quality metric. The normalized MSE, $\text{NMSE}_{l,k} = \text{tr}(C_{l,k}) / \text{tr}(R_{l,k})$, quantifies estimation quality (0 = perfect, 1 = useless) and is the natural new figure for the estimation paper.

Who does it this way / references. This is the canonical cell-free LMMSE channel estimator of Demir-Bjornson-Sanguinetti [2, Sec. 4] and Bjornson-Sanguinetti [3]; the pilot-contamination treatment traces to Ngo et al. [1]. **This part of your code is correct and standard for the far field.**

4. Estimated CSI: the near-field case (the real work)

Why the far-field estimator is not enough. The LMMSE formulas above are correct for any correlation matrix R , but two near-field-specific issues arise:

- **The angular (DFT) basis fails.** In the far field a channel is sparse in the angle domain, which classical estimators exploit. In the near field the response depends on angle AND distance, so the DFT codebook is mismatched and angle-only estimators lose accuracy [5].
- **The correlation model changes.** $R_{l,k}$ must be the near-field (spherical-wave) correlation, not the planar one; using the wrong R biases the estimate.

The established near-field solution — polar-domain estimation. Cui and Dai [5] introduce a polar-domain dictionary sampled jointly in angle and distance; the near-field channel is sparse in that dictionary, and it is recovered by sparse methods (e.g., polar-domain orthogonal matching pursuit). This is the reference method for 'near-field vs far-field channel estimation' and the natural estimator to cite and adapt. A lower-complexity route that fits your code with minimal change is the **near-field LMMSE**: keep equations (Step 2) but build $R_{l,k}$ from the NUSW model so the estimator is matched to the spherical wavefront.

For the cell-free / distributed setting specifically, near-field combining and processing for cell-free XL-MIMO is treated by Wang et al. [6] and the hybrid near-far activity-detection line by Lei et al. [7]; the near-field XL-MIMO tutorial [4] is the umbrella reference. These justify doing estimation with a near-field-aware model in a cell-free system.

What your current code does — and the gap. For far-field links the code runs the correct LMMSE estimator (Section 3). For near-field links it does NOT estimate: it overwrites the estimate with the exact deterministic steering vector ($\hat{h} = h_{\text{det_all}}$) and assigns a tiny floor covariance. That is effectively PERFECT near-field CSI, not estimation. So today the code is a hybrid: near-field perfect + far-field estimated. For a channel-estimation paper this must be replaced by a real

near-field estimator (polar-domain [5] or NF-LMMSE) so that the estimated-CSI curves are genuinely estimated everywhere.

5. What to compute for the new paper

Present three clearly-separated CSI settings, run on the SAME system and the SAME (cross-AP) detector, so the paper is about acquisition, not detection:

- **Perfect CSI (upper bound):** true $h_{\{l,k\}}$ everywhere, $C = 0$. Matches your paper’s stated assumption; use it as the benchmark curve. [2,3]
- **Proposed near-field estimation:** LMMSE with the NUSW correlation (or polar-domain OMP) on near-field links, standard LMMSE on far-field links, with realistic pilots and contamination. [5,6]
- **Far-field-mismatched estimation (ablation):** estimate every link with the planar-wave model. This shows the penalty of ignoring the near field and is the ablation that motivates the proposed estimator. [5]

New figures: (a) NMSE vs SNR for the three settings; (b) BER and per-user SE vs SNR, perfect vs proposed vs mismatched, for the four detectors including cross-AP list-SIC and the IDD scheme; (c) BER vs pilot length τ_p (the estimation-overhead / contamination trade-off). Together these make ‘impact of channel estimation’ a self-contained contribution distinct from the detector paper.

Fix the inconsistency. The submitted paper states perfect CSI but its results were generated with estimated (contaminated) CSI. In the new paper, label every curve by its CSI setting and generate perfect-CSI curves with $C = 0$ and true h , and estimated-CSI curves with the estimator of Section 4 — never mix.

6. Which reference justifies which choice

Choice	Why / where it comes from	Ref
Perfect CSI as an upper bound	Standard genie benchmark isolating the detector from acquisition	[2],[3]
Far-field LMMSE pilot estimation	Canonical cell-free estimator; error covariance C , orthogonality	[2],[3]
Pilot contamination model	Co-pilot sharing corrupts the estimate; fundamental limit	[1],[2]
Near-field NUSW channel model	Spherical wavefront, amplitude+non-linear phase across the array	[4],[5]
Near-field (polar-domain) estimation	Angle-distance dictionary; DFT basis fails in the near field	[5]
Near-field cell-free processing	Distributed combining / estimation in near-field cell-free XL-MIMO	[6],[7]

Bottom line. Perfect CSI is a one-line assumption backed by [2,3] and is honest as an upper bound. Estimated CSI in the far field is the standard LMMSE of [2] — correct in your code. Estimated CSI in the near field is the genuine new work: adopt a near-field-aware estimator [5] (polar-domain or NUSW-LMMSE) in place of the current deterministic shortcut, and the perfect-vs-estimated comparison then becomes a defensible conference paper on channel estimation for near-field cell-free XL-MIMO.

References

[1] H. Q. Ngo, A. Ashikhmin, H. Yang, E. G. Larsson, T. L. Marzetta, “Cell-free massive MIMO versus small cells,” *IEEE Trans. Wireless Commun.*, 16(3):1834–1850, 2017.

[2] O. T. Demir, E. Bjornson, L. Sanguinetti, “Foundations of user-centric cell-free massive MIMO,” *Found. Trends Signal Process.*, 14(3-4):162–472, 2021.

[3] E. Bjornson, L. Sanguinetti, “Scalable cell-free massive MIMO systems,” *IEEE Trans. Commun.*, 68(7):4247–4261, 2020.

[4] H. Lu, Y. Zeng, *et al.*, “A tutorial on near-field XL-MIMO communications toward 6G,” *IEEE Commun. Surveys Tuts.*, 26(4):2213–2257, 2024.

[5] M. Cui, L. Dai, “Channel estimation for extremely large-scale MIMO: Far-field or near-field?,” *IEEE Trans. Commun.*, 70(4):2663–2677, 2022.

[6] Z. Wang, J. Zhang, *et al.*, “Low-complexity distributed combining design for near-field cell-free XL-MIMO systems,” *IEEE Trans. Wireless Commun.*, 25:11799–11815, 2026.

[7] J. Lei, *et al.*, "A unified distributed algorithm for hybrid near-far field activity detection in cell-free massive MIMO," IEEE Trans. Wireless Commun., 2026.

Prepared as supporting material for the follow-up paper on channel estimation for the cross-AP list-based SIC / IDD near-field cell-free XL-MIMO system. Equations follow the model and notation of the submitted paper; references [1]–[7] are from that paper's bibliography.